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Network and System Admin

DHCP config & Failover, Task Automation with PS (backup), ADC (FSMO role transfer)

Overview:

1.DHCP Configuration & Failover:

DHCP automatically assigns IP addresses to network devices. I have synchronized two servers with failover configuration so that if one server is down, the other one keeps the service running.

2. Backup Task Automation (PowerShell)

In the case of my 2nd task -First, I installed the 'Windows Server Backup' feature. Then I created a backup script using the wbadmin backup command. I set the 'Task Scheduler' to run the backup at a specific time every day daily at 10:00 PM. I verified using the get-backup command to confirm successful backup completion.

3. ADC & FSMO Role Transfer

In the case of my 3rd task- First, I set up an Additional Domain Controller (ADC) to ensure fault tolerance and load distribution in the network. Then, I transferred FSMO (Flexible Single Master Operation) roles from the main server to the ADC server to maintain Active Directory continuity.

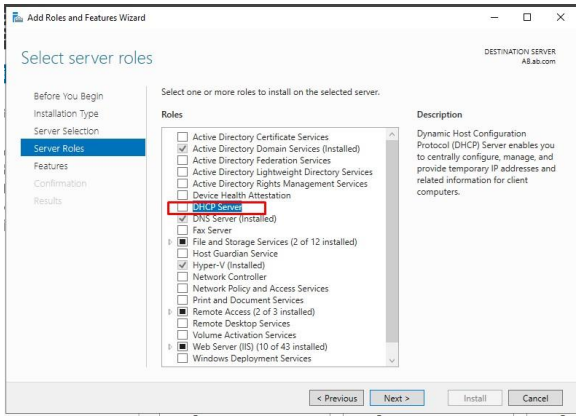
1. <u>DHCP Configuration & Failover</u>

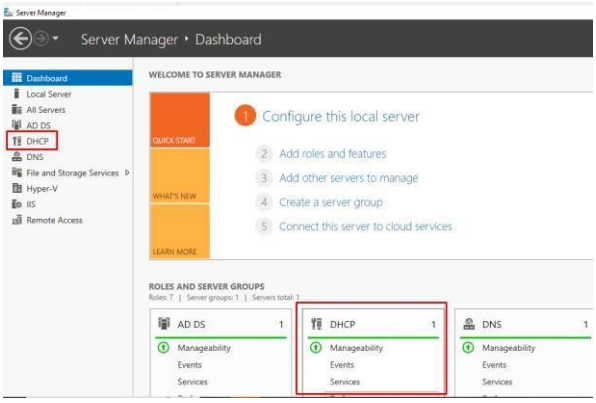
Purpose: To automate IP address assignment and ensure network availability.

Use Case: Essential for eliminating manual configuration errors and preventing network downtime if a primary server fails.

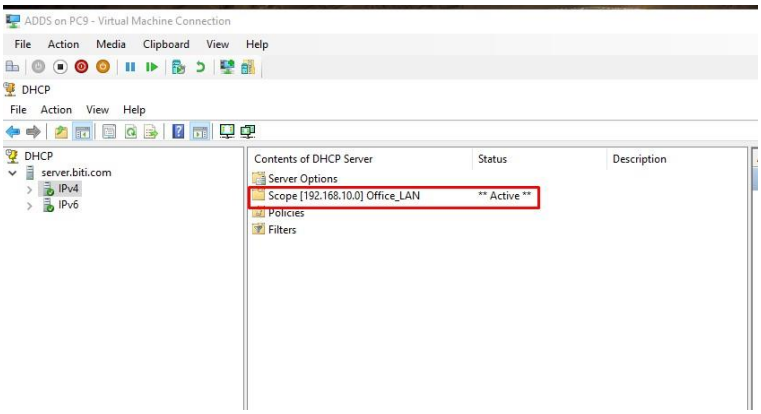
Mechanism: It follows the DORA (Discover, Offer, Request, Acknowledge) process. Windows Server 2022 supports Load Balance (50/50 distribution) and Hot Standby (active/passive) modes**.**

Implementation Steps: Install DHCP Role: *Server Manager* > *Add Roles and Features*.



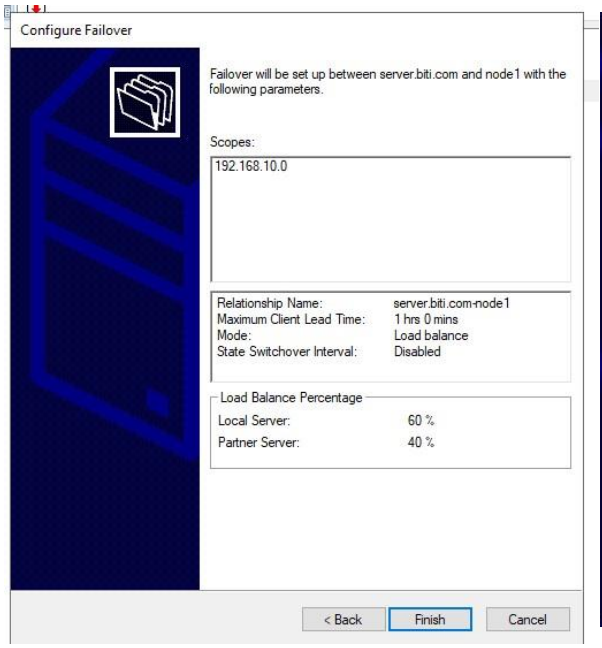


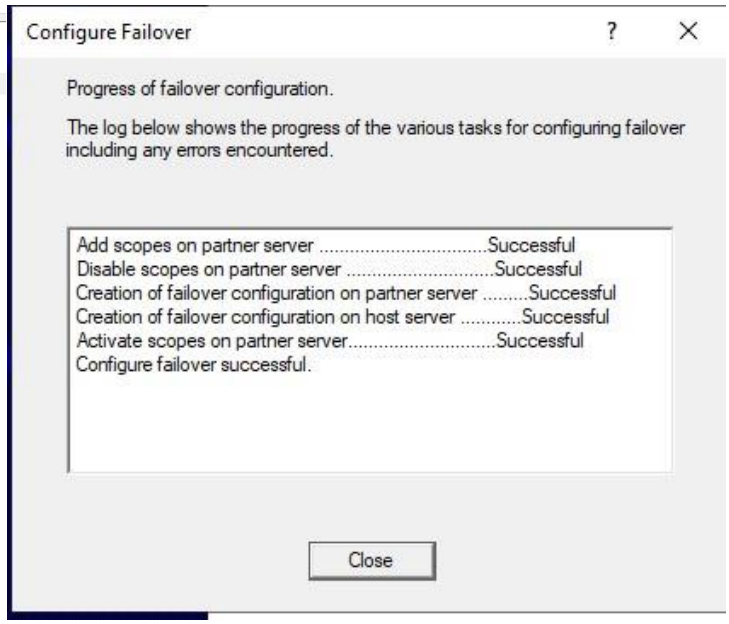
Create Scope: Right-click IPv4 > *New Scope*. I have set the IP range and subnet mask.

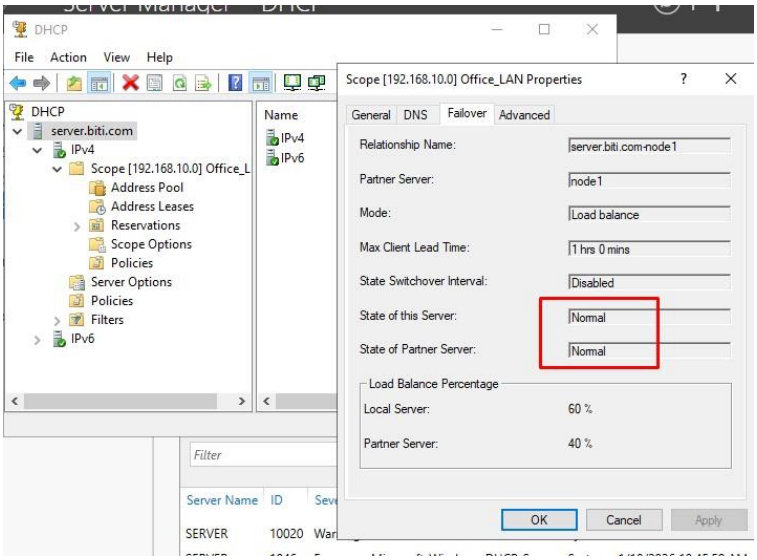


Configure Failover:

- Right-clicked the scope > Configure Failover.
- Selected the partner server.
- Selected the mode (Load Balance).
- Set a shared secret (password) for authentication.

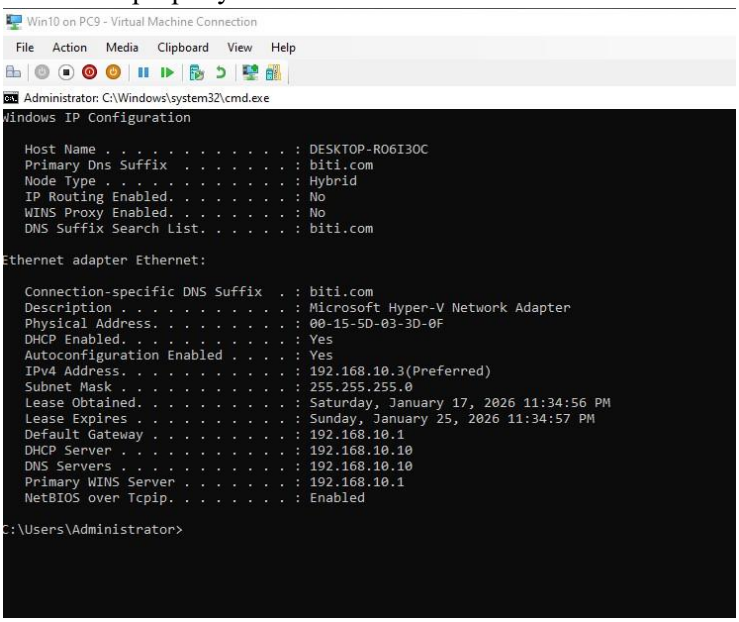


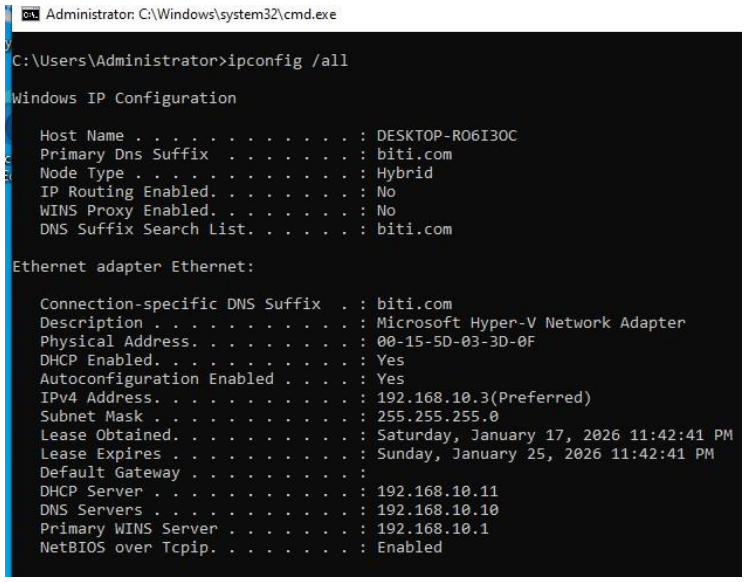




Here it is seen that the state of the Server and State of Partner Server are normal. That is, Failover is Successful. In addition, the load balance here is 60% on the Local server and 40% on the Partner Server.

below, the local server is turned off on the client PC to show whether the partner server is able to provide the service properly.





```
Administrator: C:\Windows\system32\cmd.exe
C:\Users\Administrator>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-RO6I3OC
   Primary Dns Suffix  . . . . . . . : biti.com
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : biti.com

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : biti.com
   Description . . . . . . . . . . . : Microsoft Hyper-V Network Adapter
   Physical Address. . . . . . . . . : 00-15-5D-03-3D-0F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.10.3(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Saturday, January 17, 2026 11:42:41 PM
   Lease Expires . . . . . . . . . . : Sunday, January 25, 2026 11:42:41 PM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.10.11
   DNS Servers . . . . . . . . . . . : 192.168.10.10
   Primary WINS Server . . . . . . . : 192.168.10.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

2. **<u>Backup Task Automation (PowerShell)</u> Purpose:** To eliminate manual intervention in data protection Automatic data backup via script.

Why use it?

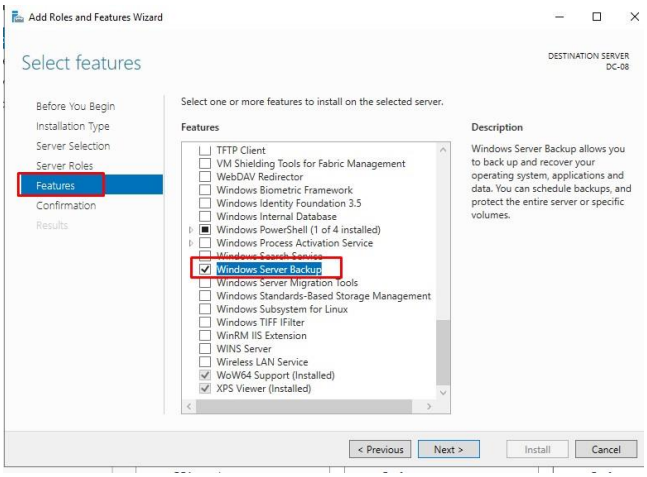
- **Reliability:** Backups run without human intervention.
- **Customization:** The entry can script specific parts of the system state. • **Resource Management:** I changed it to 10:00 PM due to server downtime during news hours**.**

Use Cases:

- Automated system state backup
- Server configuration backups

Mechanism: PowerShell script is created using the WindowsServerBackup module. The command is then run via powershell.exe at a specific time (e.g-10:00 PM) using Task Scheduler.

PowerShell Backup Workflow: Install
Windows Server Backup feature-



Create and Run PowerShell script for backup

```
Administrator: Windows PowerShell
C:\Users\Administrator> wbadmin Start backup -backupTarget:Z: -include:C: -allCritical -Quiet
dmin 1.0 - Backup command-line tool
 Copyright Microsoft Corporation. All rights reserved.

rieving volume information...
s will back up System Reserved (100.00 MB),(C:),(\\?\Volume{4cb52d8b-0000-0000-0000-205f11000000}\) to Z:
 backup storage location that you selected is on the same physical hard
k as your operating system. If the hard disk fails, you may lose both
r system data and backup data.

 backup operation to Z: is starting.
ating a shadow copy of the volumes specified for backup...
ating a shadow copy of the volumes specified for backup...
dows Server Backup is updating the existing backup to remove files that have
n deleted from your server since the last backup.
s might take a few minutes.
 backup of volume System Reserved (100.00 MB) completed successfully.
ating a backup of volume (C:), copied (2%).
ating a backup of volume (C:), copied (11%).
ating a backup of volume (C:), copied (21%).
ating a backup of volume (C:), copied (28%).
ating a backup of volume (C:), copied (36%).
ating a backup of volume (C:), copied (43%).
ating a backup of volume (C:), copied (51%).
ating a backup of volume (C:), copied (61%).
ating a backup of volume (C:), copied (69%).
ating a backup of volume (C:), copied (79%).
ating a backup of volume (C:), copied (88%).
ating a backup of volume (C:), copied (96%).
ating a backup of volume (C:), copied (100%).
```

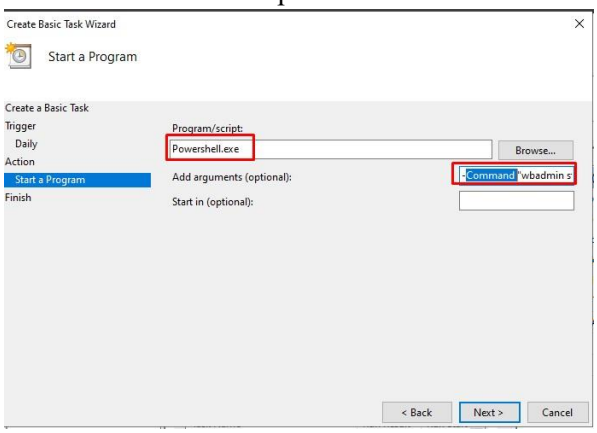
I backed up the entire drive using the one line above.

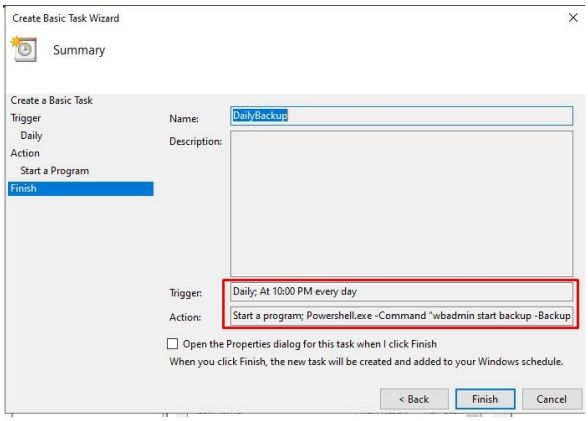
```
he backup operation successfully completed.
he backup of volume System Reserved (100.00 MB) completed successfully.
he backup of volume (C:) completed successfully.
he backup of volume  (524.00 MB) completed successfully.
:\Windows\Logs\WindowsServerBackup\Backup-18-01-2026_10-20-12.log

S C:\Users\Administrator>
```

Automation Rules (Task Scheduler):

I followed the steps below to automatically run the command at a specific time every day:

- I opened Task Scheduler.
- Then I clicked on **Create Basic Task**.
- Here I gave a name "DailyBackup" and set the *Daily* **trigger**.
- I selected *Start a Program* as the **Action.**
- In the Program/script box I wrote: *powershell.exe*
- In the **Add arguments** box I wrote the following code.
- The file named powershell.exe is stored on the Z drive.

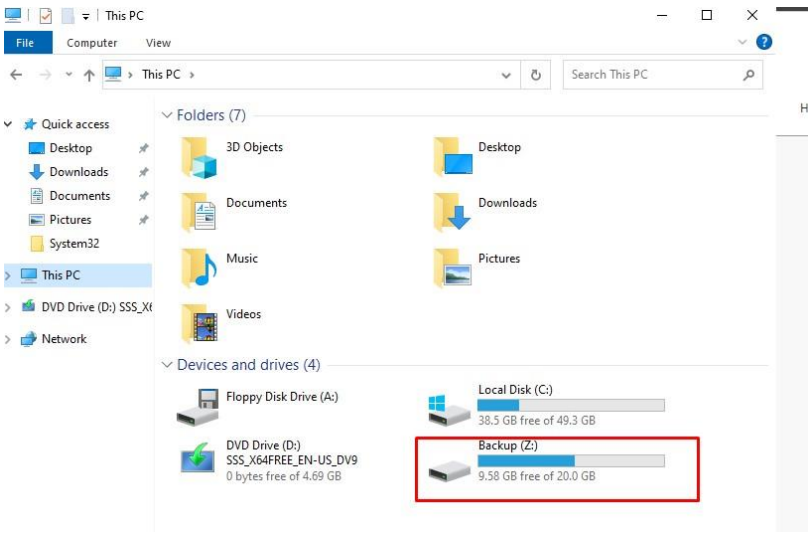




Command to check backup status: To check if the backup is going properly or not,
I gave the following short command.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-WBJob -previous 1

JobType          : Backup
StartTime        : 1/18/2026 4:20 PM
EndTime          : 1/18/2026 4:23 PM
JobState         : Completed
CurrentOperation :
HResult          : 0
DetailedHResult  : 0
ErrorDescription :
JobItems         : {VolumeList, BareMetalRecovery}
VersionId        : 01/18/2026-10:20
SuccessLogPath   : C:\Windows\Logs\WindowsServerBackup\Backup-18-01-2026_10-20-12.log
FailureLogPath   : C:\Windows\Logs\WindowsServerBackup\Backup_Error-18-01-2026_10-20-12.log


PS C:\Users\Administrator>
```

The backup has been successfully completed.



Benefits of Automated Backup: Ensures regular and accurate backups without human intervention.

3. <u>Additional Domain Controller (ADC)</u>

An ADC is a secondary controller that keeps a replica of the Active Directory database.

Why use it?

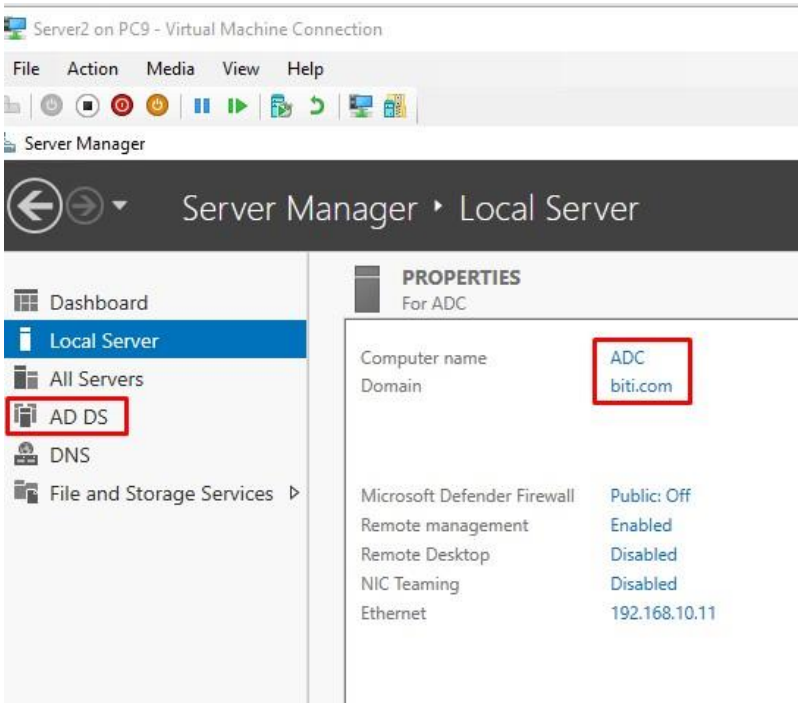
- **Fault Tolerance:** Keeps the network running if the main DC fails**.**
- **Load Balancing:** Spreads out login traffic. • **Speed:** Helps users in remote offices log in faster locally.

Mechanism: Uses Multi-master Replication via the KCC to keep all DCs synced.

Implementation Steps: • Server Manager > Add Roles and Features.

- In Server Manager, click the **Notification Flag** and select **Promote this server to a domain controller**.
- **Deployment Configuration:** Select **"Add a domain controller to an existing domain"**.

- In the **Domain** box, type: biti.com. • Click **Select** to provide Domain Administrator credentials biti\Administrator.

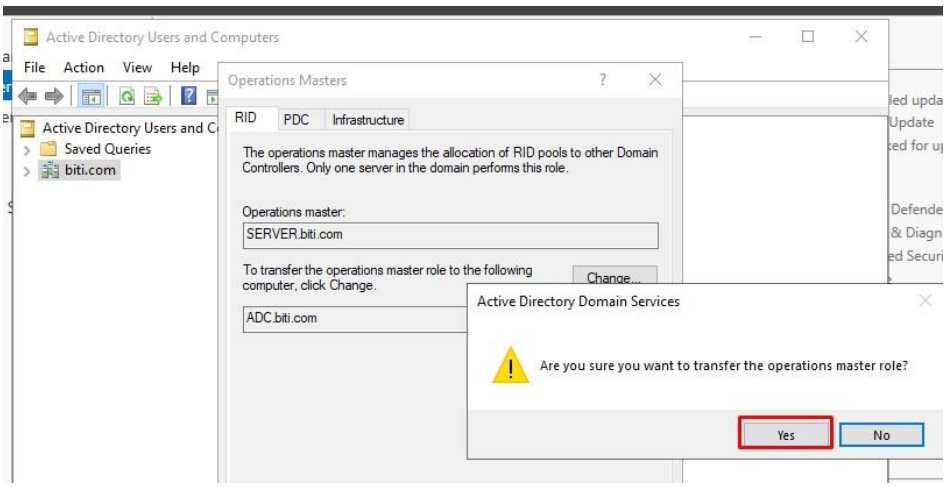


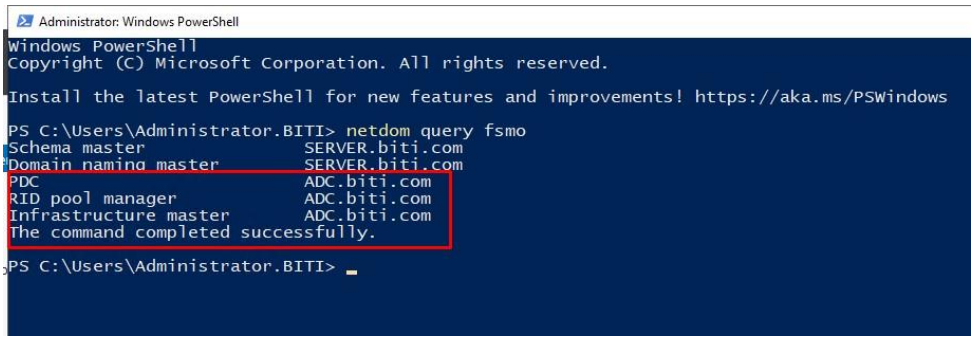
4. FSMO Role Transfer (GUI) • **Use Cases:** Server upgrades, maintenance, or better load distribution.

Implementation Steps:

A. Transferring RID, PDC, and Infrastructure Roles:

1. Opened Active Directory Users and Computers.
2. Then right-clicked on your domain name and selected Operations Masters.
3. Now I see three tabs (RID, PDC, Infrastructure).
4. Now I have changed 3 Roles RID, PDC and Insfrastructure Roles step by step.





I have completed FSMO Role Transferred.

Thank You